

Maurice D. Hanisch

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Education

California Institute of Technology, Ph.D. Math & CS ○ Pursuing research in ML for quantum chemistry and materials science, advised by Prof. Anima Anandkumar. ○ Fully funded through the Kortschak Fellowship, awarded based on academic excellence and research potential.	Current GPA: 3.94	09/2025 – Present Pasadena, USA
ETH Zurich, M.Sc. Physics ○ Expanded analytical skills with a focus on quantum theory and information technologies, with 30% excess credits in mathematics, ML and probabilistic AI involving 8 team projects, all graded 6.00/6.00 (🟢 projects).	5.77/6.00 (Top 12%)	09/2022 – 05/2024 Zurich, Switzerland
LMU Munich, B.Sc. Physics ○ Acquired extensive grounding in key physics fields, providing a strong foundation in the principles driving modern technologies, and fostering advanced analytical and problem-solving skills.	1.28/1.00 (Top 10%)	10/2018 – 09/2022 Munich, Germany

Work Experience

Bloomberg CTO Office, Security Research Intern, Julien Vanegue's Team ○ Stress-tested the CHERI CPU architecture, a \$300 million industry-academic security effort, by attempting to break its memory-safety guarantees to determine its suitability for Bloomberg's infrastructure. ○ Discovered multiple attack vectors that compromise both spatial and temporal memory safety guarantees. ○ Contributed mitigations to the open-source CHERI software via multiple PRs (🟢 jemalloc, 🟢 dlmalloc, 🟢 c-guide). ○ Co-authored a paper accepted at ACM CCS 2026.		06/2025 – 09/2025 New York, USA
California Institute of Technology, ML Research Intern, Anima Anandkumar's Group ○ Led a large-scale benchmark, retraining seven SOTA molecular models for open-shell and charged systems, validating our proposed method, <i>OrbitAll</i> , in our published work. ○ Developed a differentiable quantum feature generation pipeline for GNN-based models, enabling end-to-end differentiability of physics-informed learning of quantum chemical properties.		09/2024 – 05/2025 Pasadena, USA

Research Experience

IBM, Master's Thesis, Stefan Woerner's Group 📄 Thesis (PDF), 🌐 Project repository ○ Conducted state-of-the-art research on error correction at a global leader in the quantum computing industry. ○ Developed a Python/C++ decoding pipeline that scaled to billions of measurement records per run, enabling one of IBM's then-largest error-correction experiments. ○ Designed and deployed the full experimental pipeline on IBM Q hardware, showing that analog-information decoding can improve logical error rates by up to 30x in large-scale repetition-code experiments.	6.00/6.00	09/2023 – 05/2024 Zurich, Switzerland
ETH Zurich, Summer Project, Jonathan Home's Group 📄 Thesis (PDF) ○ Investigated the motional interaction of ions trapped in an electromagnetic field for use in quantum computing. ○ Designed and simulated trapping potentials using Python to enable a beamsplitter interaction. ○ Experimentally validated the designed potentials on GKP-encoded qubits within a trapped-ion setup.		06/2023 – 08/2023 Zurich, Switzerland
MPI for Quantum Optics, Bachelor's Thesis, Ignacio Cirac's Group 📄 Thesis (PDF) ○ Investigated efficient algorithm design and resource optimization for Gaussian quantum circuits. ○ Applied complexity theory and mathematical techniques to prove that bosonic Gaussian circuits can be implemented on quantum computers with exponentially fewer qubits compared to a direct encoding.	1.30/1.00	04/2022 – 09/2022 Garching, Germany

Selected Publications

Learning the Kohn-Sham map with neural operators for quasi-linear scaling density functional theory ○ D. Khan*, <u>M.D. Hanisch</u> *, N. Argatoff, E. Xie, S. Sharma, A. Anandkumar - arXiv:2608.23895 (2026)	
Soft information decoding with superconducting qubits ○ <u>M.D. Hanisch</u> , B. Hetényi, J.R. Wootton - APS Open Sci. 1, 000019 (2026)	
MGB: The Material Generation Benchmark ○ L. Yan, B.S. Kang, <u>M.D. Hanisch</u> , J. Ma, A. Anandkumar - NeurIPS AI4Mat Workshop (2025)	
OrbitAll: A Unified Quantum Mechanical Representation Deep Learning Framework for All Molecular Systems ○ B.S. Kang, V.C. Bhethanabotla, A. Tavakoli, <u>M.D. Hanisch</u> , ..., A. Anandkumar - arXiv:2507.03853 (2025)	

Extracurricular Activities and Competitions

○ Participated in 10+ technical competitions and workshops across cities like Paris, Munich, Zurich and New York, sponsored by companies ranging from consulting firms (Bain, McKinsey) to tech giants (Bloomberg, Infineon, NVIDIA, EDF).
○ Frequently selected through competitive application processes, often with expenses (travel, accommodation, meals) covered.
○ Won over \$3500 in hackathon prizes for projects using recurrent NNs, time series, LLMs, CUDA-Q, SageMaker and Hugging Face.
Hackathons: 1 st /30 at ETH QHack 2024: 🟢 NVIDIA task 1 st /12 at LOQC 2023: 🟢 EDF task 2 nd /6 at ETH QHack 2023: 🟢 IQM task

Technical Skills and Interests

Programming:	Python, C, C++, Fortran, Matlab, SQL PyTorch, TensorFlow, Scikit-learn Git, Linux, Latex, Docker
Languages:	German (native), French (native), English (C1: IELTS 8/9), Italian (A1)
Interests:	Weightlifting, Volleyball, Calisthenics, Surfing, Chess